Category:

Steganography

Name:

Cat Tail

Message:

It holds not one, but three hidden messages. The only clue we have is a note saying, “cats’ tails tell a lot.” Can you reveal the mystery and find all the flags?

Provided file: cat_tail.png

Hint:

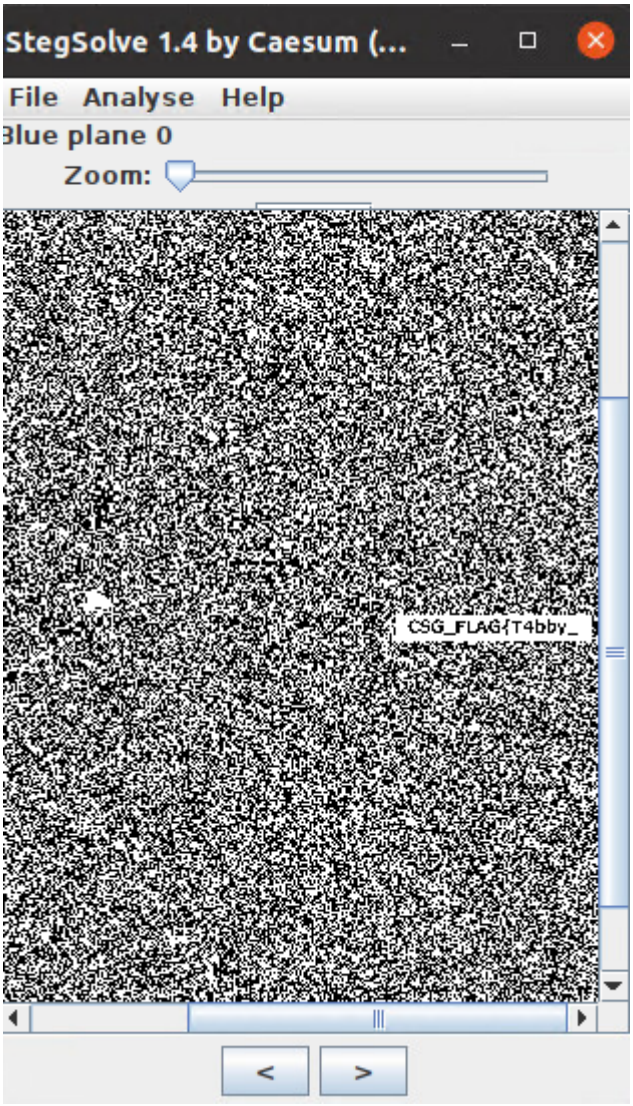
- You can interpret “tail” in multiple ways (the cat’s actual tail, the end of the file, details)!

Instructions:

1. Opening the provided file, you will see a cat image.



2. The note in the challenge description suggests we should pay attention to the cat’s tail. You won’t find any hint no matter how significantly you would zoom in. In such case, Least Significant Bit (LSB) steganography is a strong suspect. Open any LSB analysis tool (e.g. StegSolve) to investigate further. Upon inspecting the Blue channel’s 0-bit plan (LSB), the first flag fragment emerges (“CSG_FLAG{T4bby_”} around the cat tail area.



3. Next, check the file metadata using tools such as ExifTool. In one of the shown tags, you will find the second fragment of the flag.

```
chen@ubuntu:~/Desktop$ exiftool cat_tail.png | grep TAIL
Comment                         : deTAIL: Second flag fragment is 'Wisp3rs_Myst3ri0us_'
```

4. There is another fragment to be found. The word “tail” in the challenge description could also refer to the end of the file’s binary data. The standard PNG End Of Image chunk marker is IEND (hex: 49 45 4E 44). However, if you look at the binary data, this file is appended with flag fragment directly after the EOI marker.

```
0075a7d0: 6729 0000 0000 4945 4e44 ae42 6082 4330  g).....IEND.B`.C0
0075a7e0: 6433 735f 772f 5f54 6169 6c7d 0a         d3s_w/_Tail}.
```

Flag is:

CSG_FLAG{T4bby_Wisp3rs_Myst3ri0us_C0d3s_w/_Tail}